

Topic: Cartesian Product Operation

Q. What is a Cartesian Product in DBMS? Explain its working mechanism and provide an example demonstrating how it combines two relations.

****Definition to Remember****

The ****C**
row of the
columns

Cartesian Product Operation (x)

* ****Binary operation**** - operates on two tables (which need NOT be union compatible).

* ****Sy**

2. Working of Cartesian Product

* Takes ****every single row from A**** and pairs it with ****every single row from B****.

* Gene

A x B :

Row1_1

3. Example

COLORS (2 rows)

| Color_ID | Color_Name |

| :--- | :-

SHAPES (2 rows)

| Shape_ID | Shape_Name |

| :--- | :-

Query: COLORS x SHAPES

Output:

| Color_ID | Color_Name | Shape_ID | Shape_Name |

| :--- | :-

*****Must-Write Points (for 10 marks)*****

1. Carte

*****Quick Recall*****

`x (Cart
Founda